Digital payment adoption across Indian states, 2018–2026: building the state panel from PhonePe Pulse

BTP study, Part 1

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Abstract

1 Introduction

1.1 The question and the data gap

This study is Part 1 of a B.Tech project on digital payments and financial inclusion across the states of India. The project asks whether the states that adopted digital payments faster also became more financially inclusive and grew faster. It also asks in which direction the relation goes. Adoption can increase inclusion, inclusion can make adoption faster, or both can occur. The measure of adoption is the use of the Unified Payments Interface (UPI). UPI is the real-time payment system that the National Payments Corporation of India (NPCI) operates.

These questions need data on adoption for each state over many periods. Data for one period show which states have more adoption. They cannot show whether a change in adoption comes before or after a change in inclusion. The methods of the project use the changes within each state over time: fixed-effects panel regressions and tests of the direction of the relation. These methods need a state panel, that is, one value for each state in each period. The states also differ in population, so the panel must divide each total by the population of the state.

NPCI publishes UPI statistics, but its data by state start only in April 2025 [3, 1]. A panel from the NPCI data thus has no quarter before April 2025. It cannot show how adoption changed in the years before 2025. The project therefore needs another source with data by state over a longer period.

1.2 PhonePe Pulse as a proxy

PhonePe is a digital payment app that uses UPI. The company publishes PhonePe Pulse, an open data repository with transaction counts and amounts, and user counts, for each state and each quarter from 2018Q1 [4]. This study uses PhonePe Pulse as the measure of adoption. The data cover the payments of one app, not all UPI payments. Part 2 of the project compares the PhonePe values with the NPCI state data for the quarters that both sources cover.

Reddy et al. [5] used the state and district data of PhonePe Pulse to study how UPI spread across India. They found that the regions with high early adoption stayed ahead, with little change in the ranks of states and districts. Their paper uses an earlier release of the data. The release that this study uses restates all quarters with new definitions (Section 2.1). Thus their numbers and the numbers in this study are not comparable.

1.3 Aim and contents of this part

The aim of this part is to build the state-quarter panel of PhonePe adoption and its population denominator, to check it, and to describe it. The panel has 35 state units and 34 quarters, from 2018Q1 to 2026Q2. Statistics across states use a main sample of 30 units. All values in this part are nominal: the study does not adjust the amounts for changes in prices.

Section 2 describes the data sources and the files that the study uses. Section 3 explains how the raw files become the panel: the state units, the parsing rules, the data checks, the quarterly population and the measures. Section 4 describes adoption in India as a whole. Section 5 compares the states. Section 6 gives the limitations of the data, and Section 7 gives the conclusions. Appendix A lists the state units, and Appendix B gives the commands that make all the numbers again. Part 2 of the project converts the amounts to real values with consumer price indices. It also compares PhonePe Pulse with the NPCI state data and shows the data on maps. Part 3 tests whether the states with low adoption move closer to the leading states over time (convergence).

2 Data sources

The panel has data from PhonePe Pulse for adoption and from the population projections of the National Commission on Population (NCP 2020) for the denominator. This section describes each source and the files that the study uses.

2.1 PhonePe Pulse

PhonePe Pulse is a public Git repository on GitHub. In it, PhonePe publishes aggregate and anonymised counts of the transactions and users of its app [4]. The licence is the Community Data License Agreement – Permissive, Version 2.0 (CDLA-Permissive-2.0). This licence lets any person use, change and share the data, if the licence text goes with the shared data. PhonePe adds new data to the repository each quarter. Thus the study uses the data at one fixed commit: `257bf38`, of 24 September 2026. All PhonePe values in this study come from this commit.

The `data` folder of the commit has the sections `aggregated`, `map` and `top`. Each section has folders for transactions, users and merchants. The `aggregated` files give totals for India and for each state, with the transactions split by type. The `map` files give values by state (the "hover" files) and by district. The `top` files list the states and districts with the largest values. Each JSON file holds one quarter. For example, the file `2021/1.json` has the data for January to March 2021.

The commit is a new release of the data. Its README says that PhonePe restated the earlier quarters, from January–March 2018, with new definitions of the metrics. The README also advises users not to compare figures from this release with figures that PhonePe published before. It advises users not to join the data of this release and of earlier releases into one series. This study follows that rule: it uses only the files of the pinned commit.

Table 1 lists the files that the study uses, with their content and the number of items.

The transaction hover file of a quarter has one entry for each state and union territory. Each entry gives the number of transactions in the quarter (the count) and their total value (the amount). The README does not state the unit of the amount. This study reads it as nominal rupees. The user hover file gives the number of registered users of each state in the quarter. The README does not say whether this number is cumulative. The number never falls from one quarter to the next in any state (Section 3.3). This result agrees with a cumulative count of registrations. Each hover file has 36 raw names. The interim file `phonepe.csv` has 3,570 rows: one row for each state id, quarter and variable.

The aggregated state files split the count of each state by type. The README defines the types as follows. P2P transactions are person-to-person money transfers. Utility transactions

Table 1: PhonePe Pulse files used in the study. Unit: state-quarter. Sample: 36 raw state and union-territory names, 34 quarters, 2018Q1–2026Q2. Source: PhonePe [4], pinned commit.

Folder (under `data/`)	Content and use	Items
`map/transaction/hover`	Transaction count and amount by state	34 quarter files; 3,570 rows in the interim file
`map/user/hover`	Registered users by state	34 quarter files; in the same interim file
`aggregated/transaction/country/india/state`	Transaction count by type: P2P, Retail, Utility	36 names × 34 quarters; 3,570 rows in the interim file
`aggregated/transaction/country` and `aggregated/user/country`	National totals, used for checks only	34 quarter files each

are recharges and bill payments. Business transactions are payments to merchants. This type replaces the earlier types for financial services, merchants and others. The files use the label "Retail" for the business type, and this study uses the file labels. The README says that the types do not overlap and that they sum to the total count. The type files give counts only, with no amounts. Thus the study can split the count by type, but not the amount. The interim file `phonepe_types.csv` has 3,570 rows: one row for each state id, quarter and type. The national files of the `aggregated` section give the total count and users of India. The study uses them only to check the state files (Section 3.3).

2.2 Files not used

The study does not use the merchant files. The README defines registered merchants as the life-till-date count of unique merchants on the platform. In each quarter from 2018Q1 to 2019Q2, one or more raw units have no merchant file. A panel of merchants would need values for these gaps, and the data do not give them.

The study does not use the district files or the top lists. The unit of the project is the state. District data would need a separate crosswalk of district names, and this study does not make one. The top lists give only the units with the largest values in a quarter, so they do not cover all units.

The pinned commit has no files on insurance, on app opens or on the device brands of users. The README still describes the user data as "broken down by devices". But the user files at this commit give only the registered count.

2.3 NCP 2020 population projections

The denominator is the projected population from the report of the Technical Group on Population Projections. The National Commission on Population issued this report, and the cover of the report gives the date November 2019 [2]. The report projects the population of India and of each state and union territory for each year from 2011 to 2036. For the states except Goa, and for Delhi and the union territory of Jammu and Kashmir, the report uses the component method. This method projects fertility, mortality and migration from the 2011 Census and the Sample Registration System. The report applies it to the north-eastern states other than Assam as one group. For Ladakh, the report uses the difference between its projections for the former state and the union territory of Jammu and Kashmir. For Goa and the other union territories, it uses a mathematical method. This method fits exponential growth rates by least squares to the Census growth rates from 1981 to 2011.

The study uses Table 8 of the report. This table gives the projected total population, in thousands of persons, on 1 March of each year. The study uses the years 2018 to 2027. In each year,

the table has 37 rows for the states and union territories and a row for India. A Python module reads the table from the PDF of the report into the file `data/manual/population_ncp2020.csv`. Section 3.4 explains how the annual values become quarterly values, and why the panel needs the year 2027.

2.4 Data flow

Figure 1 shows the path of the data from the raw files to the outputs of the study. The top row has the raw sources. The second row has the Python modules that read them. Each module writes interim files in a long format, with one row for each state id, period and variable. The module `parse_phonepe_types` also writes the file of data checks. The R script `part1_study.R` reads the interim files. It writes all the figures, the tables and the numbers of this study. The text gets each number from the file `numbers.tex` that the script writes, so the text has no hand-typed data value (Appendix B).

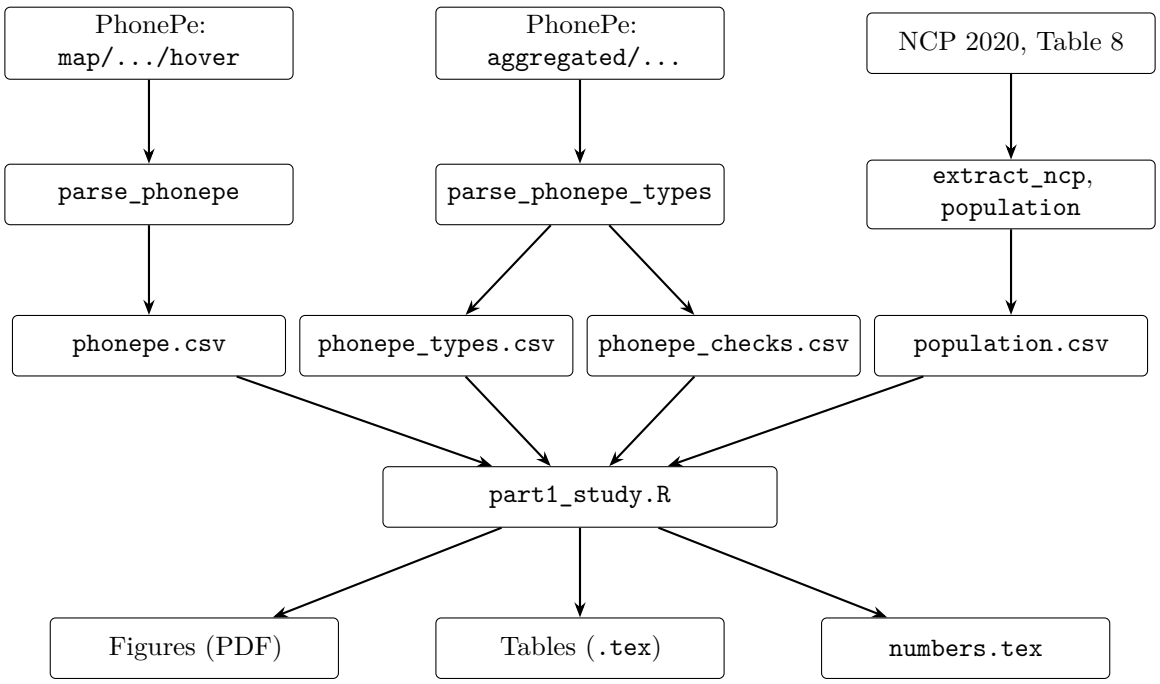


Figure 1: Data flow from the raw sources to the study outputs. Raw files: PhonePe [4] and National Commission on Population [2]. Top row: raw sources; second row: Python modules; third row: interim files; then the R script and its outputs. All interim files except the check file have the columns `state_id`, `period`, `variable`, `value` and `source`.

3 Construction of the panel

This section explains how the raw files become the panel. It describes the state units, the parsing rules, the data checks, the quarterly population and the measures.

3.1 State units and the crosswalk

The sources do not use the same names for the states. For example, NCP 2020 writes "NCT of Delhi" and PhonePe Pulse writes "delhi". A crosswalk file gives the state id for each name variant. Before the lookup, a function changes the name to lower case, replaces "&" with "and" and removes extra spaces. The crosswalk has 46 name variants and 35 state ids. It also has old

names, for example "Orissa" for Odisha. If a source has a name with no row in the crosswalk, the parse stops. Thus no state can drop out of the panel without an error.

The sources do not divide the country in the same way. PhonePe Pulse gives Jammu and Kashmir and Ladakh as separate units, and NCP 2020 does the same. NCP 2020 gives Dadra and Nagar Haveli and Daman and Diu in separate rows, but PhonePe Pulse gives them as one unit. A state id must cover the same area in all sources. Thus the crosswalk puts Ladakh into the id of Jammu and Kashmir. It also puts Dadra and Nagar Haveli and Daman and Diu into one id. Section 3.2 gives the rule that sums the parts of these merged ids.

The panel has 35 state ids. The main sample has 30 of them: the states, Delhi, and Jammu and Kashmir with Ladakh. The other 5 ids are the union territories of Andaman and Nicobar Islands, Chandigarh, Dadra and Nagar Haveli and Daman and Diu, Lakshadweep, and Puducherry. The figures and tables show all 35 ids and mark these 5 union territories. Statistics across states, such as variances, rank correlations and the variance decomposition, use only the main sample. This sample rule is the same in all parts of the project. Shares of the national total use all 35 ids, because the national total is their sum. Appendix A lists the units with their raw names in each source. The two-letter codes in the figures follow ISO 3166-2:IN [6].

3.2 Parsing and the summation rule

Python modules read the JSON files of each quarter. The PhonePe module takes the state name, the count and the amount from the transaction hover file. It takes the registered users from the user hover file. The type module reads the aggregated file of each state folder. A folder name uses hyphens in place of spaces, for example `jammu-&-kashmir`. The module changes each hyphen to a space and then finds the state id in the crosswalk. The modules do not round the values.

A merged id gets the sum of its parts in each quarter. The rule applies to all additive variables: count, amount, users, type counts and population. Before the sum, the rule finds all the parts that a source gives for a merged id over the full period. It then checks that each quarter has all these parts. If a part is absent in a quarter, the parse stops. This check prevents a merged id from losing a part in some quarters and not in others.

The type module also compares values that the repository gives in more than one file. For each state and quarter, the sum of the type counts must equal the count in the hover file. For each quarter, the national count file must equal the sum of the state counts. This file gives the type counts of India in the `aggregated` section. The national user file must equal the sum of the state user counts. If one of these comparisons fails, the module stops and names the file.

A type file can lack the entry for a type. The module then tests the present types. If they sum to the hover count of that state and quarter, the module sets the absent type to zero. If they do not, the module stops and names the file. Thus a zero in the panel comes from the data, not from an assumption. The type files have 1 absent entry: the file of Lakshadweep for 2018Q1 has no Utility entry. Its P2P and Retail counts sum to the hover count of Lakshadweep for that quarter, so the module sets the Utility count to zero.

3.3 Data checks

Table 2 gives the results of all the data checks. Each row names the check and the unit of its items. The column "Flagged" gives the number of items that fail the check or, for the zero-fill row, the number of absent entries.

The PhonePe checks pass with no exception. In all 1,224 state-quarters, the type counts sum to the hover count. The type checks use the 36 raw units before the merge, so Ladakh and Jammu and Kashmir count as separate units. Of the 3,672 type entries, 1 is absent, and the module sets it to zero (Section 3.2). The national count file and the national user file equal the

Table 2: Data checks. Unit: the item named in each row (state-quarters, state-quarter-type entries, quarters, pairs of consecutive quarters, state rows or panel cells). Sample: the 36 raw PhonePe units for the type and national checks, the 35 state ids for the user and panel checks, and the state rows of NCP 2020, Table 8; 34 quarters, 2018Q1–2026Q2. The last column is the largest absolute difference for the PhonePe rows and the ratio of the state sum to the India row for the NCP rows. Source: PhonePe [4], National Commission on Population [2].

Check	Items	Flagged	Difference or ratio
PhonePe: sum of the three type counts against the state count (state-quarters)	1,224	0	0
PhonePe: absent type entries set to zero (state-quarter-type entries). Flagged: Lakshadweep, 2018Q1, Utility	3,672	1	0
PhonePe: national count file against the sum of the state counts (quarters)	34	0	0
PhonePe: national user file against the sum of the state user counts (quarters)	34	0	0
PhonePe: registered users lower than in the previous quarter (pairs of consecutive quarters)	1,155	0	–
NCP 2020, 2018: sum of the state rows divided by the India row (state rows)	37	–	1.0000015
NCP 2020, 2027: sum of the state rows divided by the India row (state rows)	37	–	0.9999993
Population: missing cells in the panel (35 ids × 34 quarters)	1,190	0	–

sums of the state files in all 34 quarters. Thus the state files account for the full national totals, and the panel can form national totals as the sum of the state ids.

The registered users of a state never fall from one quarter to the next. The check covers 1,155 pairs of consecutive quarters, and 0 pairs show a fall. A count of active users can fall from one quarter to the next, but a cumulative count cannot. The result thus agrees with a cumulative count of registrations. It does not prove that the count is cumulative.

The NCP rows check the population table. The sum of the 37 state rows divided by the India row is 1.0000015 in 2018 and 0.9999993 in 2027. The sum of the state rows minus the India row is 2 thousand persons in 2018 and −1 thousand persons in 2027. Table 8 rounds each row to the nearest thousand, so the sum of the rows can differ from the India row. The differences are smaller than the largest difference that this rounding can give. The last row shows that the population panel has a value for every state id and quarter.

3.4 Population by quarter

NCP 2020 gives the population on 1 March of each year, but the panel needs one value for each quarter. The study uses the 1 March value of year y as the value of the first quarter of y. For each later quarter, it adds one quarter of the change to the next year:

$$P_{y,q} = P_y + \frac{q-1}{4}\,(P_{y+1} - P_y), \qquad q = 1, \ldots, 4. \tag{1}$$

Here P_y is the population on 1 March of year y, and $P_{y,q}$ is the population in quarter q of year y. This is a linear interpolation between the 1 March values. It assumes an even change within each year. The quarters of 2026 after the first need the value for 2027, so the study uses Table 8 up to 2027.

The 35 state ids have a total population of 1,318.7 million in 2018Q1 and 1,426.0 million in 2026Q2. This is an increase of 8.1 percent. The growth differs between the states. Figure 2 shows the change in population of each state id from 2018Q1 to 2026Q2.

The figure sorts the state ids by growth, and grey bars mark the union territories outside the main sample. The largest growth is 75.5 percent, for Dadra and Nagar Haveli and Daman and

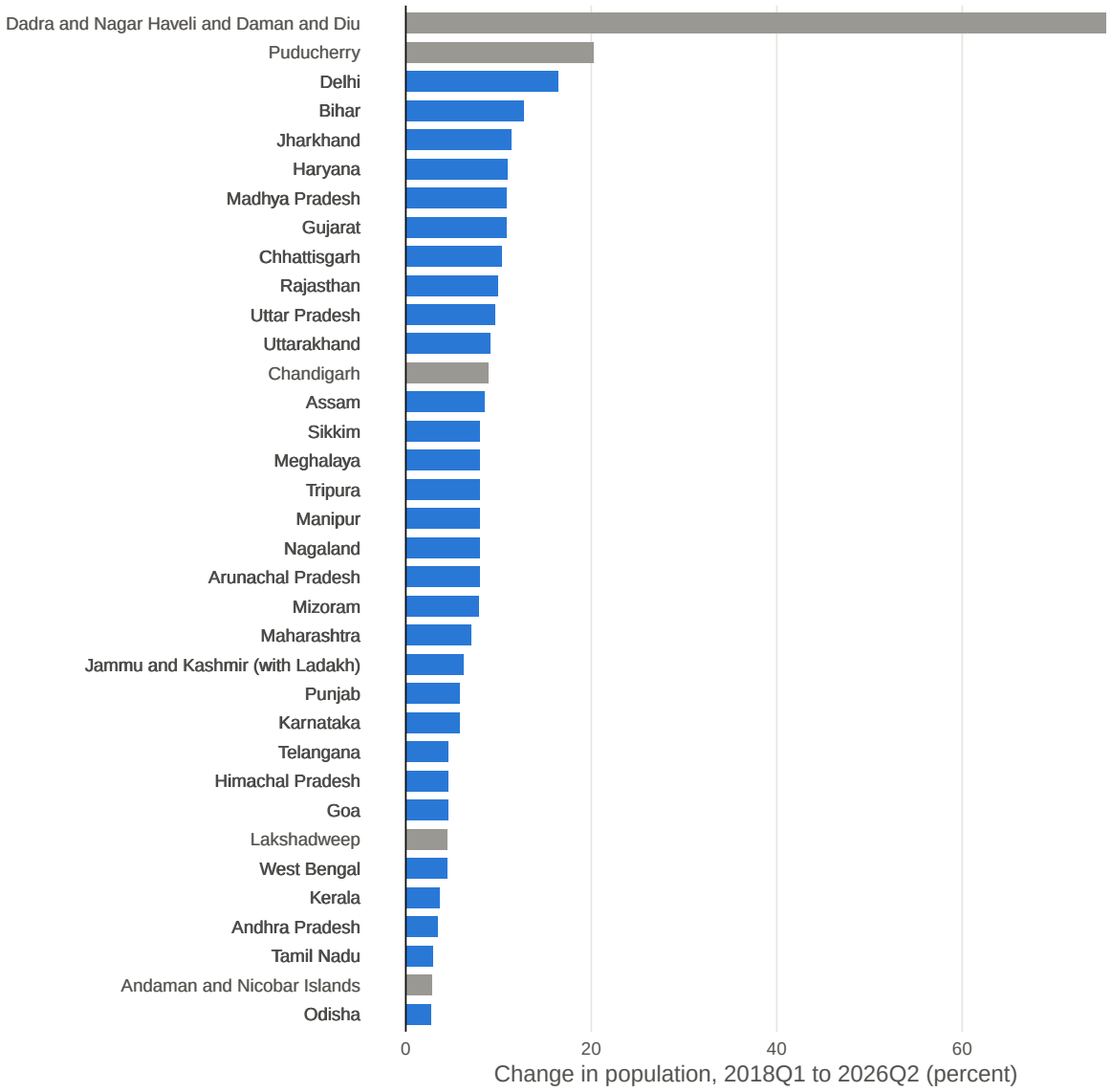


Figure 2: Population growth by state unit, 2018Q1 to 2026Q2. Unit: percent change. Sample: all 35 state units; the other union territories are marked. Source: interpolation of the NCP 2020 projections [2].

Diu. The data of this study do not show the reason for this large value. A larger population gives smaller values per person, so the per-person values of this unit depend on this projection. In the main sample, the largest growth is 16.5 percent, for Delhi. The smallest growth is 2.8 percent, for Odisha. Bihar, Jharkhand and Haryana are near the top of the main sample, and Kerala, Andhra Pradesh and Tamil Nadu are near the bottom. In a state with fast population growth, a per-person value grows more slowly than its total. Thus these differences in growth matter for the comparison of states over time.

3.5 Measures

For state s and quarter t, let C_{st} be the transaction count, A_{st} the amount in nominal rupees, U_{st} the registered users and P_{st} the population. Let C^k_{st} be the count of type k, where k is P2P, Retail or Utility. The study uses these measures (without the subscripts):

$$\begin{aligned} \text{count per person} &= C/P, & \text{amount per person} &= A/P, \\ \text{users per 1000 persons (reach)} &= 1000\,U/P, & \text{transactions per user (intensity)} &= C/U, \\ \text{average transaction value} &= A/C, & \text{type share} &= 100\,C^k/C. \end{aligned}$$

The count per person is the main measure of adoption. Its unit is transactions per person per quarter. It gives the number of PhonePe transactions in a quarter for each resident of the state. The amount per person is in rupees per person per quarter. It gives the value of these transactions for each resident. Reach is in registered users per 1000 persons. It shows how many registered users the app has for each 1000 persons in a state. It does not show how many of them used the app in the quarter. Intensity is in transactions per registered user per quarter. It shows how often a registered user pays with the app. The average transaction value is in rupees per transaction. The type share is the percent of the count of a state that is of type k. The P2P, Retail and Utility shares of a state add up to the whole count.

The growth of a variable X from quarter $t-1$ to quarter t is $100\,(X_t/X_{t-1}-1)$ percent. The annual rate of growth from a first to a last quarter is $100\,[(X_1/X_0)^{1/n}-1]$ percent. Here X_0 and X_1 are the values in the first and the last quarter, and n is the number of years between them. The study gives growth rates from 2018Q4, not from 2018Q1. The national count is 0.14 billion in 2018Q1 and 0.43 billion in 2018Q4, so rates from 2018Q1 start from a much smaller base. Also, the national retail share moves between 38.4 and 64.2 percent during 2018.

Count per person is the product of reach and intensity: $C/P = (U/P)\,(C/U)$. Let $y = \ln(C/P)$, $r = \ln(U/P)$ and $i = \ln(C/U)$. Then $y = r + i$ for each state. Across the states of the main sample in one quarter, the variance of y is thus $\text{var}(y) = \text{cov}(r, y) + \text{cov}(i, y)$. Division by $\text{var}(y)$ gives

$$1 = \frac{\text{cov}(r, y)}{\text{var}(y)} + \frac{\text{cov}(i, y)}{\text{var}(y)}. \tag{2}$$

The first term is the contribution of reach, and the second term is the contribution of intensity. Each term is the slope of a least-squares line of r (or i) on y across the states. The reach term gives the part of the differences in adoption between states that goes with differences in reach. The intensity term gives the part that goes with differences in use per user. A term can be negative, and the other term is then larger than 1. The decomposition shows how the differences divide between reach and intensity. It does not show a cause. Section 5 gives the results for each quarter.

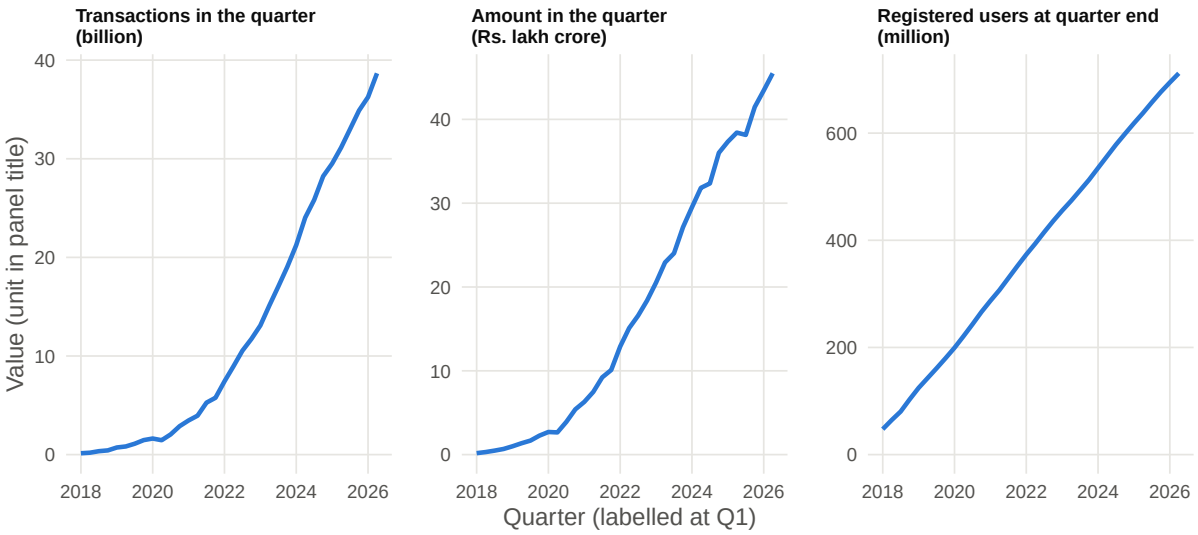


Figure 3: National transaction count (billion), transaction amount (Rs. lakh crore) and registered users (million), by quarter. Unit: national total of all 35 state units. Period: 2018Q1–2026Q2. Source: PhonePe [4].

Table 3: National annual totals and growth. Unit: count in billion, amount in Rs. lakh crore (nominal), users at year end in million, count per person per year, average value in Rs., growth of count in percent. Sample: all 35 state units. Period: calendar years 2018–2026 (2026 has two quarters). Source: PhonePe [4]; population from National Commission on Population [2].

Year	Count (billion)	Amount (Rs. lakh crore)	Users at year end (million)	Count per person	Average value (Rs.)	Growth of count (percent)
2018	1.11	1.63	103.0	0.8	1,467	–
2019	4.12	6.29	179.9	3.1	1,525	271.9
2020	8.02	14.65	266.2	5.9	1,827	94.5
2021	18.45	33.09	351.7	13.5	1,793	130.0
2022	38.64	62.97	436.0	28.0	1,630	109.4
2023	64.30	94.60	513.0	46.2	1,471	66.4
2024	99.33	129.71	598.7	70.8	1,306	54.5
2025	128.54	155.38	676.8	90.8	1,209	29.4
2026 (Q1–Q2)	74.92	88.94	711.7	52.6	1,187	–

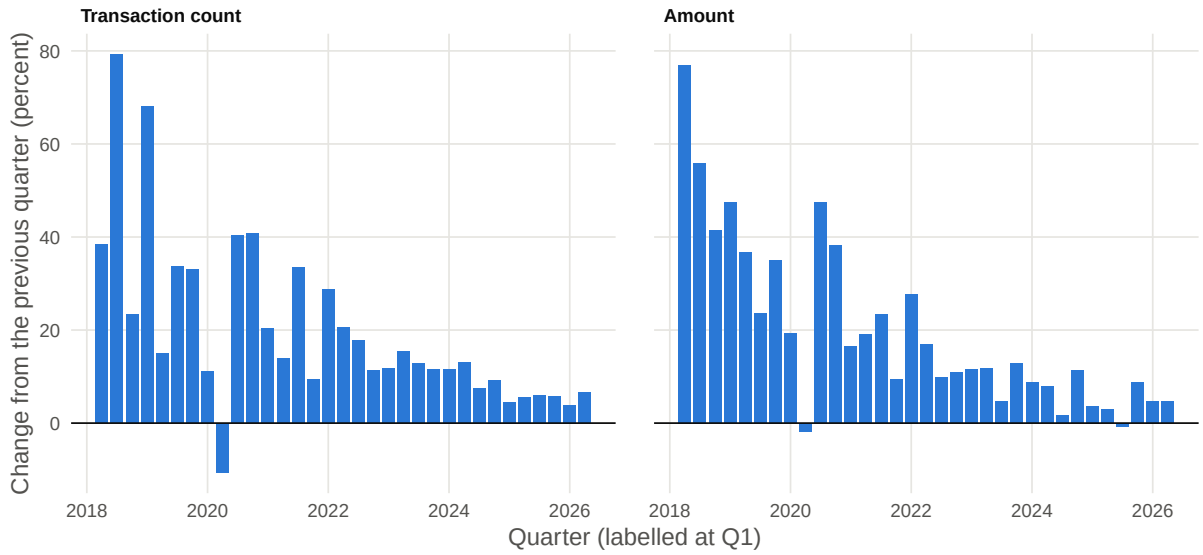


Figure 4: Quarter-on-quarter growth of the national transaction count and amount. Unit: percent change from the previous quarter. Sample: national total of all 35 state units. Period: 2018Q2–2026Q2. Source: PhonePe [4].

4 Adoption in India as a whole

4.1 Totals and growth

4.2 Quarter-on-quarter growth

4.3 Average transaction value

4.4 Transaction types

5 Adoption across states

5.1 Transactions per person

5.2 Concentration

5.3 Rank persistence

5.4 Reach and intensity

5.5 Decomposition of the variance

5.6 Retail share

6 Limitations of the data

7 Conclusions

References

[1] Chaitanya Kohli. Maharashtra leads UPI transactions as NPCI releases state-wise data. MediaNama, 14 July 2025, `https://www.medianama.com/2025/07/223-maharashtra-leads-upi-transactions-npci/`, 2025.

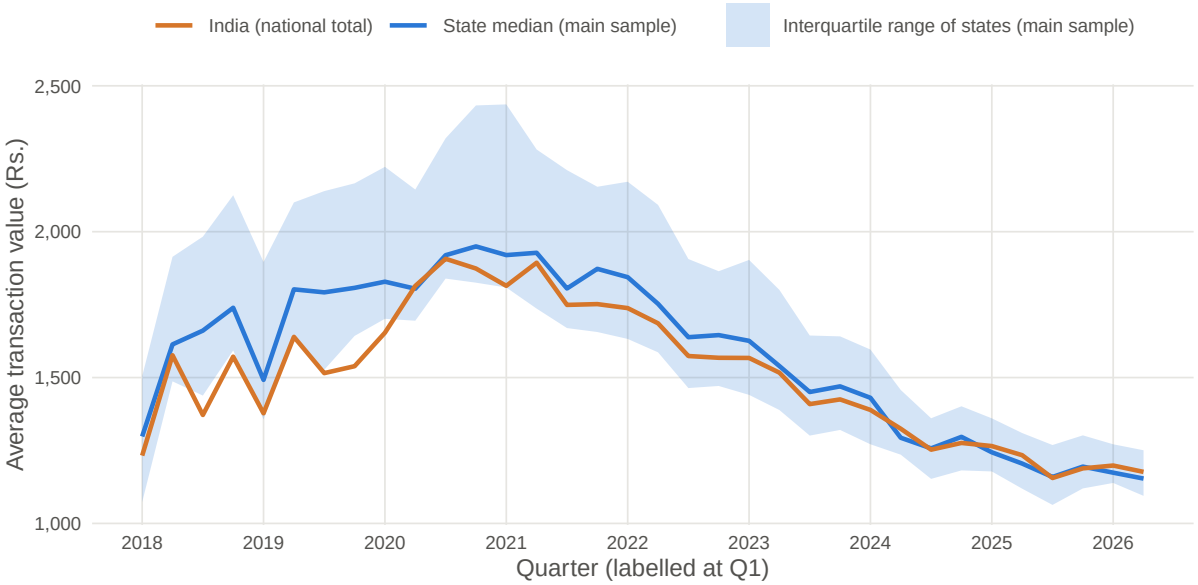


Figure 5: Average transaction value. Unit: Rs. per transaction (nominal), amount divided by count. Sample: national value, and median and interquartile range across the 30 main-sample states. Period: 2018Q1–2026Q2. Source: PhonePe [4].

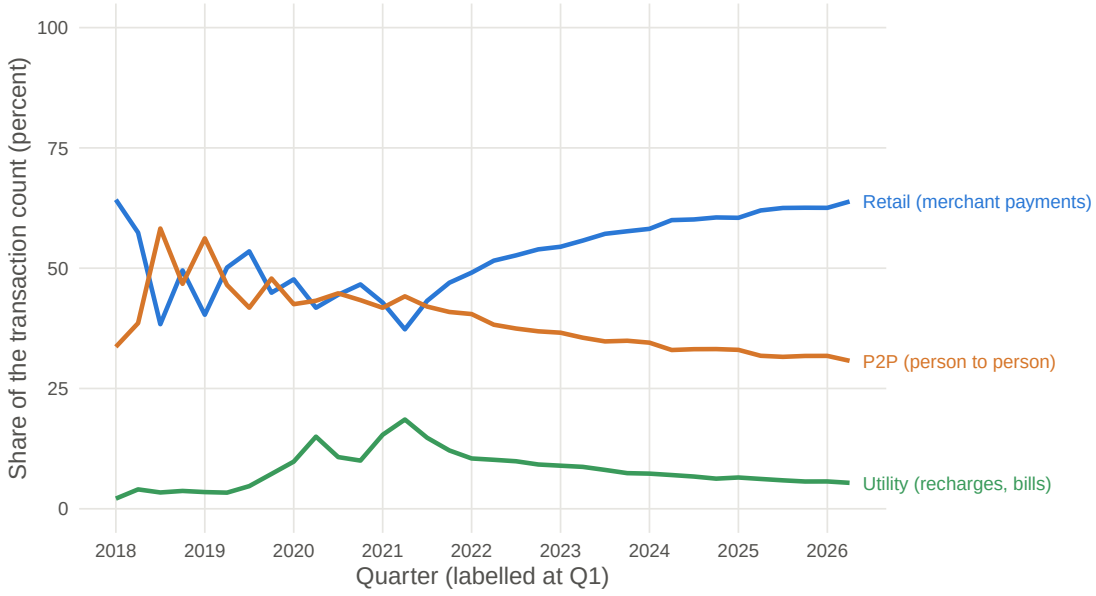


Figure 6: National shares of the transaction count by type: P2P, Retail and Utility. Unit: percent of the national count. Sample: national total of all 35 state units. Period: 2018Q1–2026Q2. Source: PhonePe [4].

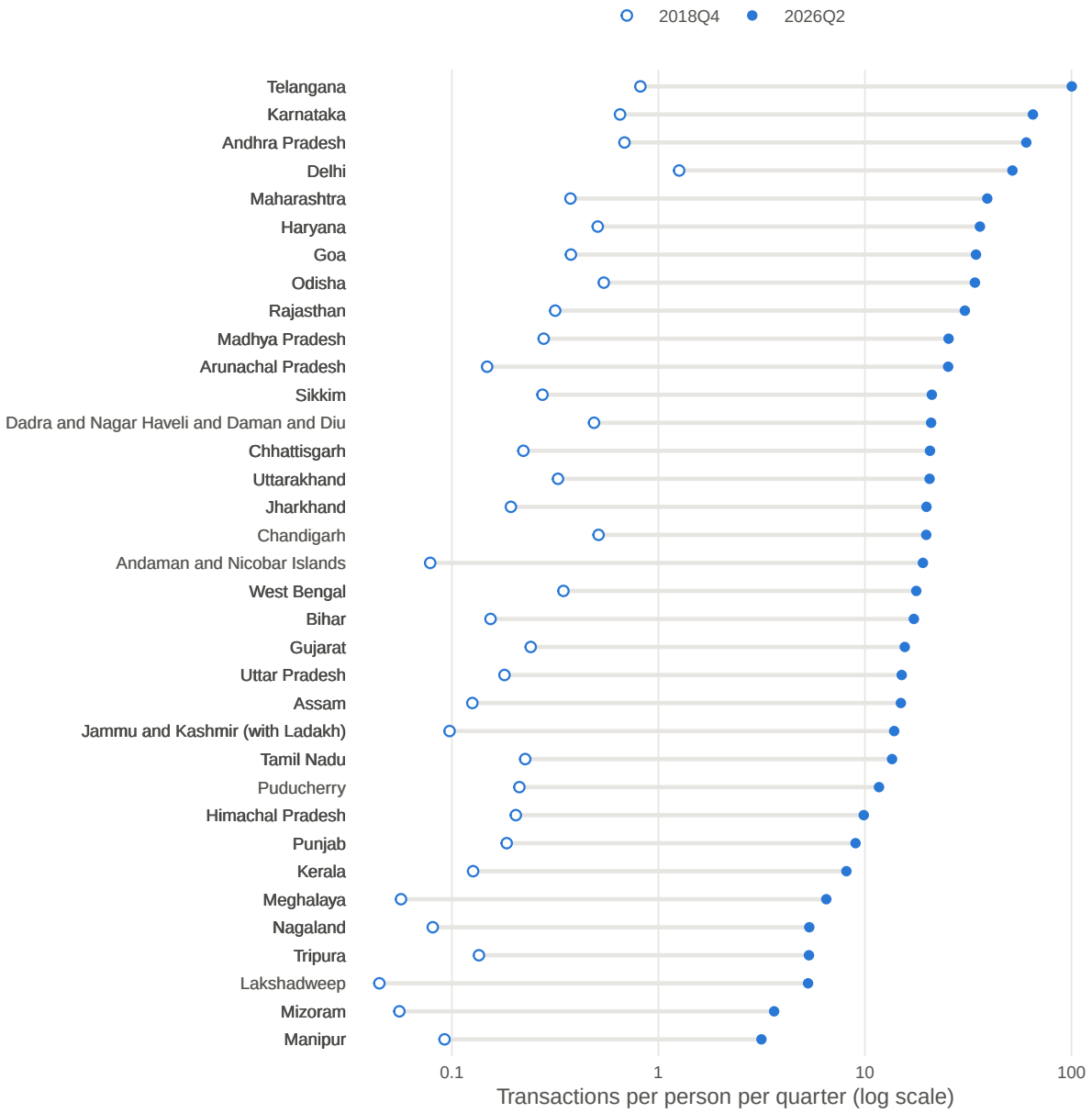


Figure 7: Transaction count per person per quarter, by state unit, in 2018Q4 and 2026Q2. Unit: transactions per person per quarter. Sample: all 35 state units; the other union territories are marked. Source: PhonePe [4]; population from National Commission on Population [2].

Table 4: State values in 2026Q2. Unit: population in million; count per person per quarter; amount per person in Rs. (nominal); users per 1000 persons; transactions per user; average value in Rs.; retail share in percent. Sample: the 30 main-sample states, sorted by count per person. Source: PhonePe [4]; population from National Commission on Population [2].

Name	Population (million)	Count per person	Amount per person (Rs.)	Users per 1000 persons	Count per user	Average value (Rs.)	Retail share (percent)
Telangana	38.66	100.3	132,629	943	106.3	1,322	66.0
Karnataka	69.05	65.1	76,649	761	85.5	1,178	67.5
Andhra Pradesh	53.73	60.4	86,051	639	94.6	1,424	61.0
Delhi	22.64	51.8	51,967	939	55.1	1,004	68.5
Maharashtra	129.52	39.1	42,197	649	60.2	1,079	67.4
Haryana	31.38	36.0	40,263	728	49.5	1,117	67.9
Goa	1.60	34.5	43,161	842	41.0	1,251	67.9
Odisha	44.69	34.1	37,283	505	67.4	1,094	61.2
Rajasthan	83.82	30.5	38,052	521	58.4	1,249	64.2
Madhya Pradesh	89.89	25.4	28,572	446	57.0	1,125	63.8
Arunachal Pradesh	1.61	25.3	29,550	592	42.7	1,169	60.8
Sikkim	0.71	21.1	25,729	574	36.7	1,221	58.3
Dadra and Nagar Haveli and Daman and Diu†	1.58	20.9	16,808	434	48.2	804	69.3
Chhattisgarh	31.29	20.7	21,684	422	49.0	1,050	62.3
Uttarakhand	12.02	20.6	21,235	562	36.6	1,033	66.4
Jharkhand	41.07	19.9	22,132	384	51.6	1,115	61.2
Chandigarh†	1.27	19.8	19,877	653	30.3	1,004	71.0
Andaman and Nicobar Islands†	0.41	19.1	25,704	543	35.1	1,347	63.4
West Bengal	100.60	17.7	21,986	430	41.2	1,241	56.4
Bihar	132.70	17.2	20,181	338	51.1	1,170	56.6
Gujarat	74.28	15.6	17,471	486	32.1	1,121	65.4
Uttar Pradesh	243.31	15.1	16,188	372	40.5	1,075	62.6
Assam	36.79	14.9	14,126	383	39.0	946	57.3
Jammu and Kashmir (with Ladakh)	14.22	13.8	15,360	415	33.4	1,109	62.3
Tamil Nadu	77.57	13.5	16,974	519	26.1	1,255	63.6
Puducherry†	1.77	11.7	13,528	409	28.6	1,157	69.1
Himachal Pradesh	7.59	9.9	10,819	474	20.8	1,096	64.1
Punjab	31.36	9.0	11,788	450	20.0	1,309	62.7
Kerala	36.23	8.1	9,260	450	18.1	1,139	65.1
Meghalaya	3.45	6.5	7,059	238	27.3	1,086	48.4
Nagaland	2.30	5.4	7,123	262	20.5	1,325	49.2
Tripura	4.27	5.4	6,265	291	18.4	1,170	56.7
Lakshadweep†	0.07	5.3	5,294	263	20.2	998	67.0
Mizoram	1.27	3.6	4,769	274	13.3	1,313	48.2
Manipur	3.32	3.2	5,118	257	12.2	1,623	43.5

† Union territory outside the main sample.

Table 5: Concentration of activity in the five states with the largest transaction count. Unit: share of the national total, in percent. Sample: all 35 state units. Periods: 2018Q4 and 2026Q2. Source: PhonePe [4]; population from National Commission on Population [2].

Share of the top five (by count)	2018Q4	2026Q2
Transaction count (percent)	46.2	52.6
Amount (percent)	44.1	53.7
Registered users (percent)	43.7	41.9
Population (percent)	42.2	37.5
States needed for half of all transactions	6	5

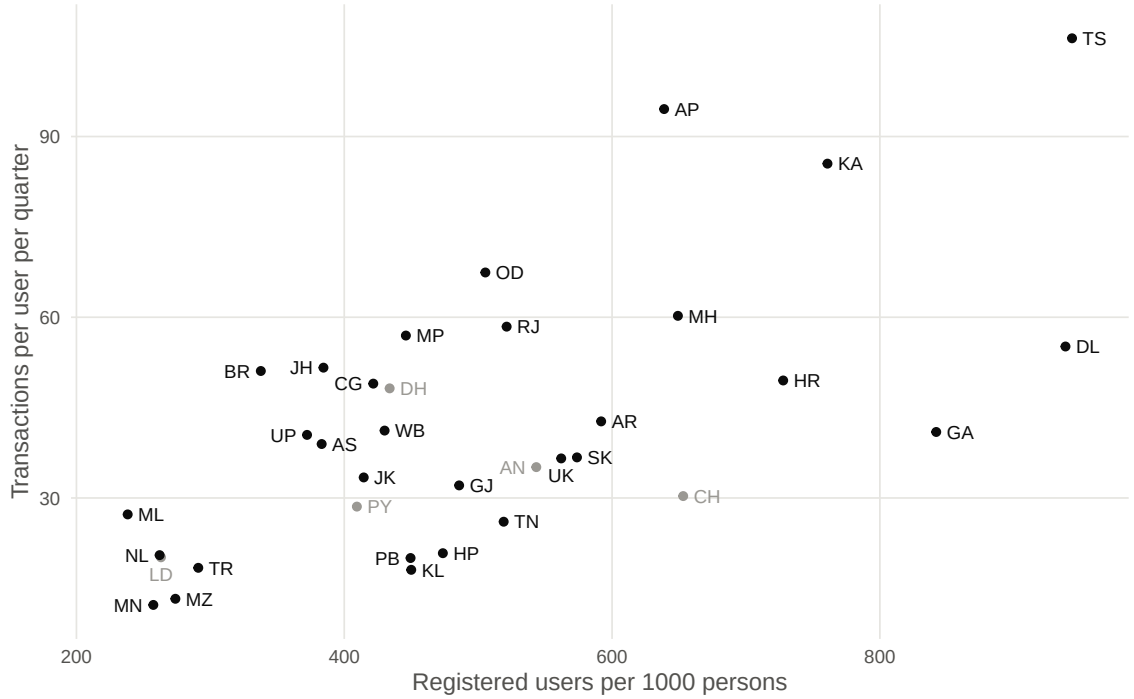


Figure 8: Reach against intensity in 2026Q2. Unit: registered users per 1000 persons (reach) and transactions per registered user per quarter (intensity). Sample: all 35 state units; the other union territories are marked. Source: PhonePe [4]; population from National Commission on Population [2].

Table 6: Variance decomposition of the log of transactions per person: $y = \ln(C/P)$, reach $r = \ln(U/P)$ and intensity $i = \ln(C/U)$. Unit: variance of logs across states; contributions as shares of $\mathrm{var}(y)$. Sample: the 30 main-sample states. Periods: 2018Q4 and 2026Q2. Source: PhonePe [4]; population from National Commission on Population [2].

	2018Q4	2026Q2
n	30	30
$\mathrm{var}(y)$	0.595	0.723
$\mathrm{var}(r)$	0.397	0.140
$\mathrm{var}(i)$	0.069	0.305
$2\,\mathrm{cov}(r, i)$	0.129	0.278
Reach contribution	0.776	0.386
Intensity contribution	0.224	0.614

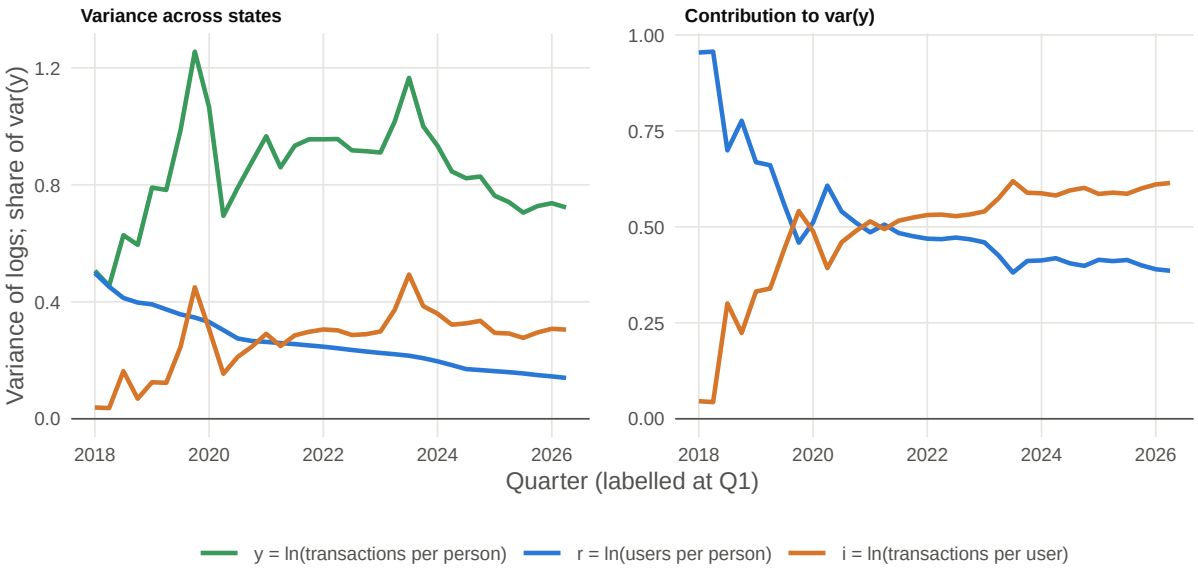


Figure 9: Contributions of reach and intensity to the cross-state variance of the log of transactions per person. Unit: share of the variance. Sample: the 30 main-sample states. Period: 2018Q1–2026Q2. Source: PhonePe [4]; population from National Commission on Population [2].

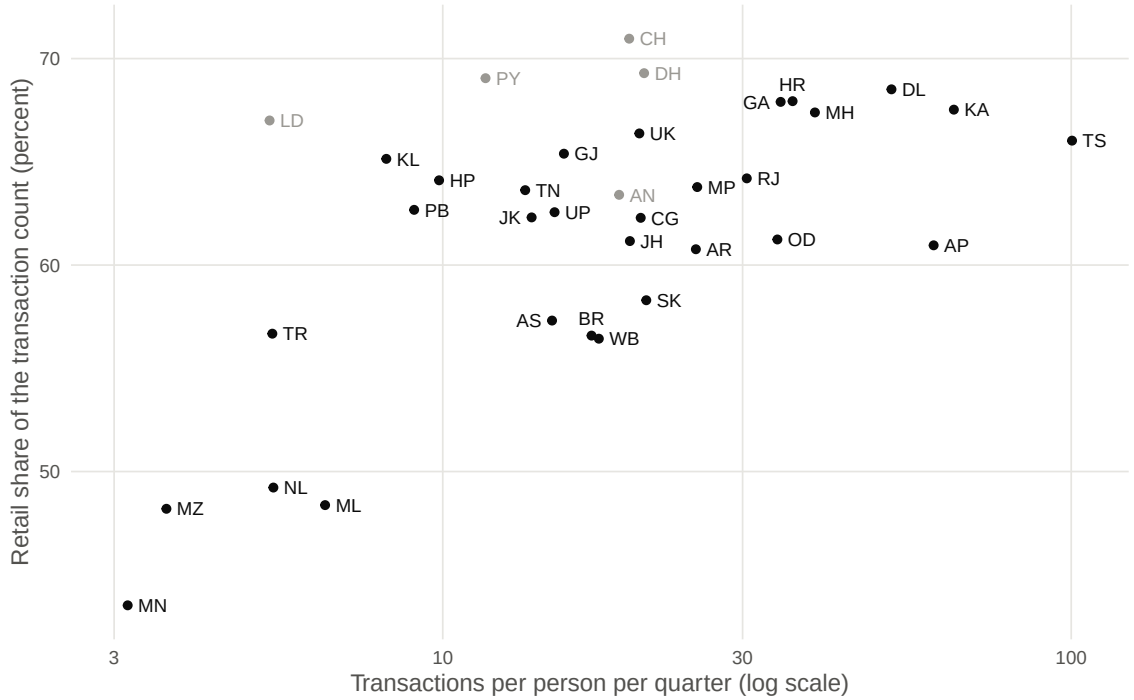


Figure 10: Retail share of the transaction count against transactions per person in 2026Q2. Unit: percent of the state count; transactions per person per quarter. Sample: the 30 main-sample states. Source: PhonePe [4]; population from National Commission on Population [2].

[2] National Commission on Population. Report of the technical group on population projections: India and states 2011–2036. Technical report, Ministry of Health and Family Welfare, Government of India, November 2019. URL https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf. Table 8, projected total population as on 1 March.

[3] National Payments Corporation of India. UPI ecosystem statistics: state-wise data. https://www.npci.org.in/what-we-do/upi/upi-ecosystem-statistics, 2026. state-wise from April 2025; accessed 1 November 2026.

[4] PhonePe. PhonePe Pulse data repository. https://github.com/PhonePe/pulse, commit 257bf38e3095020f9f516c4c72c10bdc348781b1, 2026. CDLA-Permissive-2.0; data 2018Q1–2026Q2; accessed 1 November 2026.

[5] Aarti Reddy, Mansi Kedia, and Sanjana Shukla. Diffusion of digital payments in India: Insights based on data from PhonePe Pulse. Working Paper IPCIDE-wp1, Indian Council for Research on International Economic Relations (ICRIER), March 2024. URL https://icrier.org/pdf/IPCIDE-wp1.pdf.

[6] Wikipedia. ISO 3166-2:IN. https://en.wikipedia.org/wiki/ISO_3166-2:IN, 2026. Accessed 2026-11-04.

A The state units

The names in the last two columns are the raw names after the normalisation of the state crosswalk: lower case, with "and" in place of "&". A semicolon separates the raw names that the crosswalk sums into one unit.

Table 7: The 35 state units of the panel: display name, `state_id`, two-letter code [6], membership of the main sample of 30, and the raw names that map to the unit in the PhonePe Pulse hover file for 2026Q2 [4] and in NCP 2020, Table 8 [2]. Source: state crosswalk (`state_crosswalk.csv`) and state labels (`state_labels.csv`).

Name	state_id	Code	Main	PhonePe raw names	NCP raw names
Andaman and Nicobar Islands	andaman_nicobar	AN	No	andaman and nicobar islands	andaman and nicobar islands
Andhra Pradesh	andhra_pradesh	AP	Yes	andhra pradesh	andhra pradesh
Arunachal Pradesh	arunachal_pradesh	AR	Yes	arunachal pradesh	arunachal pradesh
Assam	assam	AS	Yes	assam	assam
Bihar	bihar	BR	Yes	bihar	bihar
Chandigarh	chandigarh	CH	No	chandigarh	chandigarh
Chhattisgarh	chhattisgarh	CG	Yes	chhattisgarh	chhattisgarh
Dadra and Nagar Haveli and Daman and Diu	dnh_dd	DH	No	dadra and nagar haveli and daman and diu	dadra and nagar haveli; daman and diu
Delhi	delhi	DL	Yes	delhi	nct of delhi
Goa	goa	GA	Yes	goa	goa
Gujarat	gujarat	GJ	Yes	gujarat	gujarat
Haryana	haryana	HR	Yes	haryana	haryana
Himachal Pradesh	himachal_pradesh	HP	Yes	himachal pradesh	himachal pradesh
Jammu and Kashmir (with Ladakh)	jammu_kashmir	JK	Yes	jammu and kashmir; ladakh	jammu and kashmir; ladakh

Name	state_id	Code	Main	PhonePe raw names	NCP raw names
Jharkhand	jharkhand	JH	Yes	jharkhand	jharkhand
Karnataka	karnataka	KA	Yes	karnataka	karnataka
Kerala	kerala	KL	Yes	kerala	kerala
Lakshadweep	lakshadweep	LD	No	lakshadweep	lakshadweep
Madhya Pradesh	madhya_pradesh	MP	Yes	madhya pradesh	madhya pradesh
Maharashtra	maharashtra	MH	Yes	maharashtra	maharashtra
Manipur	manipur	MN	Yes	manipur	manipur
Meghalaya	meghalaya	ML	Yes	meghalaya	meghalaya
Mizoram	mizoram	MZ	Yes	mizoram	mizoram
Nagaland	nagaland	NL	Yes	nagaland	nagaland
Odisha	odisha	OD	Yes	odisha	odisha
Puducherry	puducherry	PY	No	puducherry	puducherry
Punjab	punjab	PB	Yes	punjab	punjab
Rajasthan	rajasthan	RJ	Yes	rajasthan	rajasthan
Sikkim	sikkim	SK	Yes	sikkim	sikkim
Tamil Nadu	tamil_nadu	TN	Yes	tamil nadu	tamil nadu
Telangana	telangana	TS	Yes	telangana	telangana
Tripura	tripura	TR	Yes	tripura	tripura
Uttar Pradesh	uttar_pradesh	UP	Yes	uttar pradesh	uttar pradesh
Uttarakhand	uttarakhand	UK	Yes	uttarakhand	uttarakhand
West Bengal	west_bengal	WB	Yes	west bengal	west bengal

B Reproduction

The commands below run from the repository root. They fetch the raw files, build the interim files, write the figures, tables and numbers, and build this document.

```
make data analysis report
```

The PhonePe Pulse files are at the commit `257bf38e3095020f9f516c4c72c10bdc348781b1`. The study reads these input files:

- `data/interim/phonepe.csv`
- `data/interim/phonepe_types.csv`
- `data/interim/phonepe_checks.csv`
- `data/interim/population.csv`
- `src/R/lib/state_labels.csv`

The script `src/R/part1_study.R` writes the figures to `output/study/part1/figures/`, the tables to `output/study/part1/tables/` and the numbers to `output/study/part1/numbers.tex`. The file `numbers.tex` is generated. Do not edit it. A number in the text comes from this file with the command `\val`, and an unknown key stops the compile.